APEX RETAIL GROUP | PRACTICAL CLASS TASKS

DOCUMENT: CLASS TASKS (SESSION 1)

SCENARIO: HR SALARY & COMPENSATION AUDIT

INSTRUCTIONS FOR LEARNER: Complete the following four tasks in your Jupyter Notebook during class. Follow the structured workflow: **Business Framing -> Code Execution -> Business Interpretation.** Do not peek at instructor solution keys until completed.

TASK 1: CLARIFYING THE BUSINESS QUESTION

Sarah Jenkins (HR Manager) asks: **"Are we paying our new 5-member analytics team fairly compared to our company benchmark of $75,000?"**

1. **Write down two specific analytical metrics you must calculate to answer her question.**
2. **Explain in one sentence why providing ONLY the total payroll sum is insufficient for Sarah's decision.**

TASK 2: CONSTRUCTING THE DATA CONTAINERS

Create a Jupyter Notebook code cell and perform the following setup:

1. Create a variable named `company_benchmark` and assign it the float value `75000.0`.
2. Create a list named `proposed_salaries` containing the five proposed team salaries: `72000.0, 81000.0, 75000.0, 69000.0, 88000.0`
3. Create a list named `employee_names` containing five string names corresponding to the team members: `"Maya", "Liam", "Sophia", "Noah", "Ethan"`
4. Print all three containers using Python's `print()` function.

TASK 3: CALCULATING PAYROLL METRICS & INDIVIDUAL VARIANCES

In a new code cell, perform the mathematical calculations:

1. Calculate total team payroll using the `sum()` function and store it in `total_payroll`.
2. Calculate the total number of employees using the `len()` function and store it in `team_size`.
3. Calculate the average salary and store it in `average_salary`.
4. Using index position access (e.g., `proposed_salaries[0]`), calculate the dollar variance between each employee's salary and the `company_benchmark`. Formula:

Individual Variance = Employee Salary - Benchmark Store these 5 variances in a new list named `salary_variances`.

5. Print `total_payroll`, `average_salary`, and `salary_variances`.

TASK 4: BUSINESS INTERPRETATION & RECOMMENDATION WRITING

Review your code output from Task 3 and answer the following questions in a markdown cell in your notebook:

1. **What is the average salary of the proposed team? Is it above or below benchmark, and by how much?**
2. **Which specific employee creates the largest positive variance above benchmark?**
3. **Draft a 2-sentence formal recommendation email response to Sarah Jenkins summarizing your findings and indicating whether the budget request is reasonable.**